

R PRO-1 PoE Dual mmWave Multisensor

PoE-powered dual-radar presence and environmental sensing for Home Assistant

The Apollo R PRO-1 is Apollo's most advanced presence sensor — a compact PoE mmWave multisensor built for high-accuracy motion detection and environmental monitoring in smart homes and commercial buildings. It carries the Made for ESPHome badge and runs fully local and open-source with no cloud or subscription.

The base sensor uses an LD2450 radar for multi-zone motion tracking; an optional LD2412 module adds excellent still-presence detection for seated or sleeping people, and the two work in tandem for comprehensive occupancy data. The R PRO-1 supports true 802.3 PoE and USB-C power, with Wi-Fi and Bluetooth built in, and ships with Wi-Fi firmware that can switch to PoE from the device dashboard.

Key Specifications

Product	R PRO-1 PoE Dual mmWave Multisensor (LD2450 optional LD2412)
Manufacturer	Apollo Automation (Versailles, KY, USA)
Primary radar	HLK-LD2450 — multi-zone motion tracking
Optional radar	HLK-LD2412 — high-accuracy still-presence detection
Microcontroller	ESP32
Firmware	ESPHome (Made for ESPHome) — open source
Power	True PoE (802.3) or USB-C
Connectivity	Wi-Fi (2.4 GHz) + Bluetooth (tracking & Improv provisioning)
Light sensing	LTR-390 — LUX + UV
Optional CO ₂	SCD40 NDIR via rear mezzanine expansion board
Dimensions	Approx. 57 x 45 mm
Onboard feedback	2 x addressable RGB LEDs + piezo buzzer
Mounting	Flat surface, single gang box, or ceiling (mounts sold separately)
In the box	Assembled sensor with case; Apollo sticker

Onboard & Optional Sensors

Sensor	Function	Availability
HLK-LD2450	24 GHz mmWave — multi-zone motion tracking with coordinates	Standard
HLK-LD2412	24 GHz mmWave — high-accuracy still-presence detection	Optional
LTR-390	Ambient light (LUX) and ultraviolet (UV)	Standard
SCD40	NDIR CO ₂ (with temp/humidity) via rear mezzanine board	Optional

The LD2450 + LD2412 combination delivers both moving-target tracking and reliable detection of stationary occupants — ideal for offices, bedrooms, and living spaces.

Configurations & Pricing

Configuration	Price (USD)
R PRO-1 (LD2450 base; LD2412 and SCD40 CO ₂ available as add-ons)	\$69.99

Optional LD2412 still-presence module and SCD40 CO₂ mezzanine are available separately. Gang-box and ceiling mounts sold separately.

Features & Use Cases

- **Presence + motion:** detect entry instantly and keep lights on while occupants sit still; place under a bed for sleep automations.
- **PoE or USB-C:** single-cable PoE install for walls and ceilings, or USB-C where preferred.
- **Light automations:** LUX triggers lights only when dark; UV monitors sunlight through windows.
- **CO₂ monitoring (optional):** NDIR SCD40 mezzanine adds accurate air-quality data and ventilation alerts.
- **Alerts:** two addressable RGB LEDs and a piezo buzzer for visual and audible notifications.
- **Modular:** rear mezzanine port supports add-on boards for future expansion.

Important Notes

- **Environmental offsets:** device heat can affect optional temperature and humidity readings and may require significant calibration offsets.
- **Recessed CO₂:** CO₂ measurements may be skewed when the sensor is recessed in a wall or ceiling.
- **Power-user tuning:** mmWave zone configuration benefits from Apollo's wiki guide and the Zone Mapper tool for inclusion/exclusion zones.
- **PoE voltage:** confirm your PoE switch settings; the R PRO-1 follows the 802.3 PoE standard.

Documentation & Resources

- **Product page:** apolloautomation.com/products/r-pro-1
- **Wiki / setup guide:** wiki.apolloautomation.com
- **Source code:** github.com/ApolloAutomation/R_PRO-1
- **Zone Mapper (beta):** github.com/ApolloAutomation/zone-mapper
- **Community:** [Apollo Automation Discord](https://discord.com/invite/apolloautomation)
- **Sensor datasheets:** Hi-Link LD2450, Hi-Link LD2412, Liteon LTR-390, Sensirion SCD40.